Hospital Patient Data Analysis and Interactive Dashboard using Python and Power BI

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1. Abstract

This project focuses on analysing hospital patient data to derive meaningful insights that can support healthcare decision-making. A publicly available healthcare dataset from Kaggle was used for the

analysis. The project involved cleaning and preprocessing raw data using Python libraries such as Pandas and NumPy, followed by exploratory data analysis (EDA) using Matplotlib and Seaborn to identify trends and patterns. The cleaned dataset was then imported into Power BI to develop an interactive dashboard featuring KPIs, charts, and slicers. The dashboard provides insights into patient demographics, treatment costs, hospital stay duration, admission types, and recovery outcomes. This project demonstrates a complete end-to-end data analytics workflow, from data preparation to business intelligence visualization.

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2. Introduction

The healthcare industry generates large volumes of patient data every day. Transforming this data into meaningful insights helps hospitals improve patient care, optimize operational efficiency, and support informed decision-making.

The objective of this project is to analyze hospital patient records and develop an interactive dashboard that highlights key healthcare metrics. Python was used for data cleaning and analysis, while Power BI was used to create an interactive dashboard that enables users to explore patient information through filters and visualizations.

This project demonstrates practical applications of data analytics techniques and business intelligence tools in the healthcare domain.

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3. Problem Statement

Hospitals collect extensive patient information, but raw data often contains inconsistencies, duplicates, missing values, and formatting issues that reduce its usefulness.

The challenge is to transform raw hospital data into a clean, structured dataset and create an interactive dashboard that enables users to monitor patient demographics, treatment costs, hospital performance, and recovery-related metrics efficiently.

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4. Objectives

The primary objectives of this project are:

Analyze hospital patient data using Python.

Perform data cleaning and preprocessing.

Handle missing values and duplicate records.

Conduct exploratory data analysis (EDA).

Visualize important healthcare trends.

Develop an interactive Power BI dashboard.

Generate business insights to support decision-making.

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5. Dataset Description

The dataset used in this project was obtained from Kaggle and contains anonymized hospital patient records.

Typical attributes include:

Attribute	Description
Patient ID	Unique identifier
Age	Patient age
Gender	Male/Female
Blood Type	Patient blood group

Attribute	Description
Medical Condition	Diagnosis
Admission Type	Emergency, Elective, etc.
Hospital	Hospital name
Doctor	Treating physician
Billing Amount	Treatment cost
Insurance Provider	Insurance details
Admission Date	Date of admission
Discharge Date	Date of discharge
Medication	Prescribed medication
Test Results	Diagnostic results

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6. Technology Stack

Tool	Purpose
Python	Data preprocessing and analysis
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Jupyter Notebook / Google Colab	Analysis environment
Power BI	Interactive dashboard development

Tool	Purpose
GitHub	Version control and project documentation

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7. Project Workflow

Dataset Collection (Kaggle)

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Data Loading

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Data Understanding

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Data Cleaning

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Data Preprocessing

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Exploratory Data Analysis

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Visualization

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Export Cleaned Dataset

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Power BI Dashboard

↓

Business Insights

8. Data Cleaning & Preprocessing

This section should describe exactly what you did. For example:

Imported the dataset using Pandas.

Examined the dataset structure using head(), info(), and describe().

Checked for duplicate records and removed them.

Identified and handled missing values.

Converted date columns into datetime format.

Standardized text values where necessary.

Exported the cleaned dataset for dashboard development.

Include screenshots of your notebook if possible.

9. Exploratory Data Analysis (EDA)

Explain the analyses you performed, such as:

Patient age distribution.

Gender distribution.

Most common medical conditions.

Distribution of admission types.

Billing amount analysis.

Hospital stay analysis.

Insurance provider distribution.

Include the corresponding charts from your notebook.

10. Data Visualization

Describe the charts you created in Python and why they are useful.
For example:

Histogram for patient age distribution.

Bar chart for medical conditions.

Pie chart for admission types.

Box plot for billing amounts.

Heatmap for correlation analysis (if applicable).

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11. Dashboard Development in Power BI

Explain that the cleaned dataset was imported into Power BI and an interactive dashboard was created.

Mention features such as:

KPI Cards

Bar Charts

Donut Charts

Line Charts

Slicers

Interactive filtering

Executive summary

Add screenshots of your dashboard.

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12. Key Business Insights

This section should summarize the main findings from **your dashboard**. Examples include:

The majority of patients belong to a specific age group.

Certain medical conditions account for a large proportion of admissions.

Treatment costs vary significantly across different conditions.

Average hospital stay differs depending on the diagnosis.

The dashboard enables users to filter data by demographics and admission type for deeper analysis.

Only include insights that are supported by your analysis and dashboard.

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13. Challenges Faced

Discuss practical challenges, such as:

Handling missing values.

Cleaning inconsistent data.

Selecting appropriate visualizations.

Designing an intuitive dashboard layout.

Ensuring the dashboard remained interactive and easy to interpret.

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14. Conclusion

Conclude by stating that the project successfully transformed raw hospital data into meaningful insights through data cleaning, exploratory analysis, and interactive visualization. Emphasize that the dashboard provides an accessible way to monitor patient information and supports informed decision-making in healthcare settings.

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15. Future Scope

Potential future improvements include:

Integration with live hospital databases.

Real-time dashboard updates.

Predictive analytics using machine learning.

Readmission risk prediction.

Hospital resource utilization forecasting.

Cloud deployment and automated ETL pipelines.

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16. References

Kaggle Healthcare Dataset

Python Documentation

Pandas Documentation

NumPy Documentation

Matplotlib Documentation

Seaborn Documentation

Microsoft Power BI Documentation